Charting the Open: Certified Reductions of Goldbach's Conjecture, and a Measurement of What They Do Not Contain

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Abstract

What can a kernel-verified system produce on a problem it cannot solve? We report a four-month campaign in which Goldbach's conjecture was written over the constructed cardinal arithmetic of an LCF kernel built on Bourbaki's own formalism, and then reduced — never proved. The campaign yields a *certified map*: eighteen kernel-judged links, replayable in a fresh process, collapsing three independent lines of attack onto a single object, so that proving the conjecture reduces to proving that for every composite k the primes below $2k$ meet their additive mirror. Two structural facts are certified about that meeting: its solutions come in pairs under the involution $m \mapsto 2k - m$, and one member of each pair lies in the lower half, so searching half the interval suffices. Our central result is negative and *measured*: re-deriving all four of the campaign's major reductions with the primality predicate replaced by a *parameter*, the same proofs close verbatim on a totally opaque set, on primality, and on a predicate for which the question is trivial. Reductions that cannot distinguish the primes from a structureless set cannot prove Goldbach — and this is a property of the statements, not a defect of the proofs. The measurement explains, at one stroke, two further paths we had already closed negatively: raw counting and equational reasoning both fail because neither ever looks at *which* integers are prime. We also report a certified fidelity defect — the repository's primality predicate did not constrain its argument to be an integer, making the formalized conjecture strictly weaker than the conjecture — proved in the kernel and repaired at provably zero cost on numerals. We claim no arithmetic progress: the conjecture is exactly as open as before. We claim that a verified system can *delimit* an open problem — draw in code, rather than by estimation, the line where the structural ends and the arithmetic begins — and that this is a mathematical object rather than a report of failure.

1 Introduction

A proof assistant is an instrument for closing goals. Pointed at an open problem it does the only thing it can: it fails, repeatedly, and the record of that failure is conventionally discarded. The question this paper asks is what such an instrument can be made to *produce* when success is excluded in advance — not as consolation, but as a mathematical result.

Our answer has three parts, and the third is the one we consider new. First, a verified system can produce a *map*: a network of certified equivalences and reductions, each link a kernel object,

*ISEP Paris. Contact: [email]. Code and certified corpus: [repository URL and tagged commit hash at submission time]. All measurements in this paper were re-run in a fresh process on 19 August 2026; the replay script is `recherche/goldbach/capstone.py`.

with the remaining obligation stated exactly at every leaf. Second, it can close paths *negatively*, and record why. Third — and this is what the map alone cannot do — it can *measure how much of the problem its own reductions actually touch*. We do this by parameterizing: the predicate "is prime" is replaced by an opaque parameter S, and the reductions are re-derived. Those that still close have been shown, by execution rather than by argument, to contain no arithmetic at all.

That measurement changes the epistemic status of a body of reduction work. A reduction that closes for every S is not a step toward Goldbach and never could have been; it is a statement about additive structure that the primes merely instantiate. Knowing which of one's results are of this kind — and knowing it mechanically, at the cost of a few seconds per instance — is the difference between a research program that is exploring and one that is circling.

The setting sharpens the experiment. We work in an LCF-style kernel implemented over Bourbaki's own formal system: the τ operator, its abbreviated assemblies, and the body of criteria Bourbaki *proves* about proofs rather than posits as rules. The arithmetic is the corpus's own constructed cardinal arithmetic, not a primitive type of naturals. Two consequences matter here. The oracle is *exact*: every claim below is a kernel object or a stated measurement, never an estimate. And because $\exists x\, \varphi(x)$ *is* $\varphi(\tau x\, \varphi)$ in Bourbaki's system rather than a separate quantifier, the conjecture can be rewritten with no existential quantifier at all — three properties of two named terms (Section 3). That is not available in a conventional setting, and it turns out to be the form in which the machine's own search organs later re-derive the statement unaided.

We report the campaign honestly, which means reporting that it did not work. Goldbach is open. Every theorem below is of the form $X \Rightarrow$ Goldbach or Goldbach $\Leftrightarrow Y$, and a test in the repository fails if any exported result ever concludes the conjecture alone (Section 8).

Contributions.

1. **A certified map of an open problem** (Section 3): eighteen kernel-judged links, replayed in a fresh process in about seven and a half minutes, under which three independent lines of attack — a bounded verification for $n \leq 86$, a reduction to composite k, and a sieve reformulation — collapse onto a single object. Proving the conjecture reduces to proving that for every composite k, the primes $\leq 2k$ meet their additive mirror.

2. **Goldbach without $\exists$** (Section 3): using Bourbaki's τ, the conjecture becomes three properties of two named terms. Both directions are certified.

3. **A measurement of arithmetic content** (Section 5, our central result): with the primality predicate as a parameter, *all four* major reductions — the sieve equivalence in both directions, the reduction to composites, the symmetry and the half-interval restriction — close verbatim on an opaque predicate, on primality, and on a trivial predicate. "Goldbach is an instance" is an execution, not a claim of prose.

4. **Two paths closed negatively, with a common cause** (Section 6): raw counting and equational strengthening both fail, and the parameterization of the previous item explains why — neither ever inspects which integers are in S.

5. **A certified fidelity defect and its repair** (Section 7): the repository's primality predicate guarded only the divisor, so the formalized conjecture quantified over non-integer witnesses and was strictly weaker than the conjecture. The defect is a kernel theorem; the repair is proved free on numerals.

6. **Two guards against the only cheating available here** (Section 8): a column separating "closed" from "axiom-free" — eleven of eighteen links are free, seven rest on the sieve's two dedicated axioms — and a test that fails if anything in the research folder concludes the conjecture.

What we do not claim. No new arithmetic fact about the primes is established. The negative results of Section 6 are measurements, not impossibility theorems: the counting result is a numerical observation about π, and the equational result is an observation about the shape of a gap produced by *our* organs at a given state. Section 5 does not show Goldbach unprovable by these methods in any logical sense; it shows that proofs which do not distinguish S from a structureless set cannot establish a statement that does depend on S. That is a delimitation, not a lower bound.

2 Background: the kernel, and an arithmetic that must be guarded

We summarize only what this paper needs; the kernel, its trust boundary, and the instrumentation around it are the subject of a companion paper. Three features of the substrate shape every result below, and each is load-bearing rather than incidental.

2.1 The trust boundary, and what "closed" means

The kernel is written in pure Python, in the LCF style: a value of type `Theoreme` is constructible only through kernel primitives, and the repository forbids fabricating one by any other route — no back-door constructor, no monkey-patching, no hand-built theorem object.

Definition 1 (Theorem object)**.** *A* theorem *is a triple $\theta = (H, C, j)$ where H is a finite set of relations (the* hypotheses *θ still depends on), C is a relation (its* conclusion*), and j names the kernel rule that produced it — including, for the axiom rule,* which theory the axiom was drawn from*. The theorem is* closed *when $H = \emptyset$.*

A *theory* in Bourbaki's sense is a name together with a finite set of explicit axioms. The reference theory of sets carries 22 of them, and that count is re-asserted at every execution of every script in this paper — it is the invariant by which the repository detects an axiom silently added anywhere.

The third component j is what makes Section 8 possible. The axiom rule produces a theorem with *empty* hypotheses: drawing A from a dedicated theory T yields a closed theorem whose dependence on T has vanished from the sequent. Closure therefore does not mean "assumption-free", and the two must be reported separately or a result rests on axioms nobody counted.

2.2 The existential quantifier is not primitive

In *Théorie des ensembles* the quantifiers are not primitive signs but *abbreviations* built from Hilbert's choice operator τ. For a relation R and a letter x,

$$(\exists x)R \;:=\; (\tau_x(R) \mid x)\, R,$$

where $(A \mid x)\, R$ denotes the assembly obtained by substituting A for every occurrence of x in R. The term $\tau_x(R)$ denotes "some object satisfying R, if there is one", so the existential statement *is* the statement that this particular named term satisfies R.

Two consequences matter here. First, an existential goal can be traded for properties of an explicitly named term, with no extra axiom — this is the mechanism of Section 3, and it is not available in a setting where $\exists$ is a primitive binder. Second, $\tau_x(R)$ contains a full copy of R, so unfolding quantifiers copies formulas into themselves and nested quantifiers grow multiplicatively; the corpus works on shared structure throughout, and the campaign's one attempt to reason on unfolded terms is reported in the companion paper as a measured failure.

2.3 Arithmetic is constructed, and cardinal — so it must be guarded

There is no primitive type of natural numbers. Integers are *finite cardinals*, and the additive laws available are those of cardinal arithmetic, developed in the book's Chapter III. This is not a stylistic choice; it is the development the repository mirrors. It has one sharp consequence, and this paper turns on it twice.

A quantifier the author believes ranges over integers in fact ranges over *cardinals* unless a finiteness guard is written. Omitting the guard is not a matter of elegance:

Remark 1 (cancellation fails without finiteness). Additive cancellation — from $a + b = a + c$ conclude $b = c$ — is *false* for infinite cardinals: $\aleph_0 + 1 = \aleph_0 + 2$ while $1 \neq 2$. A statement that uses cancellation without a Fin guard is therefore not merely unprovable, it is wrong.

Write $S^+(t) := \mathrm{Fin}(t) \wedge S(t)$ for the *guarded* form of a predicate. Section 7 reports what the missing guard cost on the statement of primality — the formalized conjecture came out strictly weaker than the conjecture — and Section 4 reports the half-interval lemma, where the guard was written from the start because the proof demanded it. The asymmetry between those two cases is the paper's most transferable observation about fidelity.

2.4 Where these results live

The repository's main tree mirrors the book's table of contents, so that an empty directory is a coverage hole and every formalized notion carries a marker naming the page and lines it transcribes. Goldbach is not in *Theory of Sets*. Results proved beyond the book therefore live in a separate `recherche/` tree, exempt from the book-anchoring discipline but subject to *identical* kernel rules: same primitives, no fabricated theorems, the 22-axiom invariant asserted throughout. A research result is not less certified — it is only off-book. Nothing in the book tree is permitted to depend on the research tree.

3 The certified map

Write H for the "halves" form of the conjecture — for every finite $k \notin \{0, 1\}$, $k+k$ is a sum of two primes — which the campaign proved interderivable with the repository's statement. The map's business is to move H to a form where what is missing is isolated at one point.

3.1 Four equivalent forms

form	statement	status	module
DEP	the repository's `goldbach()`	reference	`enonces.py`
H	halves: $(\forall k)\,\mathrm{decomp}(k{+}k)$	$\Leftrightarrow$ DEP	`enonces.py`
HC	halves, restricted to composite k	$\Leftrightarrow H$ (GG7)	`composes.py`
H_τ	canonical: no $\exists$ at all	$\Leftrightarrow H$ (GG9/GG10)	`pont_tau.py`

Composites are the whole problem. If k is prime then $2k = k + k$ decomposes for free, so the conjecture bears only on composite k. The proof is excluded middle on prime(k), with the prime branch discharged through the family $\{2p\}$ and an α-renaming bridge — the repository's statement forces two distinct spellings of `est_premier`, and the bridge crosses between them. That bridge will reappear in Section 5 as an artifact rather than a step.

The conjecture without an existential quantifier. Applying $(\exists x)\varphi \equiv \varphi(\tau x\, \varphi)$ twice, with $T := \tau p\, (\exists q \text{ mat})$ and $Q := \tau q\, (\text{mat}[p := T])$, the conjecture becomes

$$\text{prime}_1(T) \ \wedge \ \text{prime}_2(Q) \ \wedge \ 2k = T + Q,$$

three properties of two named terms. Both directions are certified and free of ad-hoc axioms. Nothing is gained mathematically — no information is created — but the *shape* of the question changes, from "does a witness exist" to "does a named term have these properties". We note in passing, since it belongs to the companion paper, that the campaign's search organs later reconstructed exactly this move on their own: a witness-*proposing* organ that fabricates $\tau x(\varphi)$ from the goal alone turns the reported gap on a decomposition goal from the intact existential statement into a statement about the properties of a named term.

3.2 The sieve, and the convergence

Let

$$P_{2k} := \{\, p : \text{prime}(p) \wedge p \in [0, 2k]\, \}, \qquad Q_b := \{\, x : (\exists y)(\text{prime}(y) \wedge b = x + y)\, \}$$

— the primes below $2k$, and the *additive mirror* of b. Both are opaque terms characterized by axioms of a dedicated theory (Section 8 accounts for them). The mirror is defined by an internal existential: no subtraction, no commutativity, no cardinality of a complement is used.

Proposition 1 (sieve equivalence, GG19, both directions)**.** $\vdash (\forall k)\big[\, (\exists m)(m \in P_{2k} \wedge m \in Q_{2k}) \iff$ $2k = p + q$ *with* p, q *prime*$\,\big]$.

Write meet(k) for the left-hand side.

Proof sketch, and where the two directions differ. The $\Leftarrow$ direction is free: under a point m of the meeting, the guarded predicate $S^{+}(m)$ falls out of the P axiom, the internal witness y (guarded, with $2k = m + y$) falls out of the Q axiom, two applications of the witness rule S5 close the double existential, and $\exists$-elimination discharges y then m. The $\Rightarrow$ direction is the one that *consumes the guard*: from $2k = p + q$ and Fin(p), the proposition "a summand of a finite sum is below it" gives $p \le 2k$; with p a cardinal and $0 \le p$ this places p in $[0, 2k]$, hence in P_{2k}; and $p \in Q_{2k}$ with internal witness q. Without Fin(p) not one of those steps is available. This is worth recording, because it is the pattern by which a fidelity defect announces itself — the guard is not decorative, the proof demands it, and Section 7 reports what happened where no proof was demanding anything.

A trap, and the discipline it forced. The mirror's defining formula binds an internal variable, and the kernel is free to α-rename that binder when it instantiates the axiom. Reconstructing the expected matrix by hand and comparing against it therefore fails intermittently and for reasons that look like nonsense. The rule adopted throughout the campaign is to *read the binder the kernel produced* off the instantiated theorem, build the supplied matrix against it, and assert equality before applying S5. Every witness-introduction site in the package does this. The same assertion caught a genuine error during the parametric port of Section 5.3: the mirror's internal witness is not in general the element being placed — it is the other summand in the equivalence, and the element itself only in the composites branch, where $2k = k + k$.

How the statements themselves are checked. A reduction is only as good as the faithfulness of the two formulas it relates, and a transcription error there would be invisible to the kernel — it would simply prove a different theorem. The package establishes fidelity by *excision*: the antecedent and the consequent are cut out of the repository's own statement rather than retyped, and the recomposition is checked for syntactic identity against it. A statement built this way cannot silently differ from the one it claims to be about; what it cannot do is guarantee that the repository's statement is itself faithful to the mathematics, which is the separate problem of Section 7.

The map's payoff is that everything the campaign proved lands on meet:

line of attack	what it had	absorbed by
bounded	Goldbach verified for all $n \leq 86$	GG25 — a concrete witness gives the meeting
composites	prime k are free	GG22 — $m := k$ witnesses
sieve	P_{2k} meets its mirror	the arrival form

Proposition 2 (synthesis, GG24)**.** $\vdash \big[\,(\forall k)\ (k\ \textit{finite, composite}) \Rightarrow \mathrm{meet}(k)\,\big] \implies H.$

Proof sketch. Excluded middle on prime(k). On the composite branch the hypothesis applies directly. On the prime branch the meeting is witnessed by $m := k$ itself: k lies in P_{2k} because it is prime and $k \leq 2k$, and in Q_{2k} with internal witness $y := k$, since $2k = k + k$ is a reflexivity. The branch then feeds the sieve equivalence and the link to the repository's historical statement. The prime branch costs one thing the abstract version of Section 5 does not pay: an α-spelling bridge, because the repository's statement of decomposition forces two distinct spellings of the primality predicate and the witness must cross between them.

Three scattered formulations became one goal. We state the obvious reservation in the repository's own words: this is an implication whose hypothesis is not proved, and that hypothesis *is* the conjecture restricted to composites. The value is in designating a single object to attack, not in having attacked it.

4 The structure of the meeting

Two facts are certified about meet. Neither says anything about existence; both constrain the shape of the solution set.

Proposition 3 (symmetry, GG23)**.** $\vdash (\forall k)(\forall m)\big[\, m \in P_{2k} \cap Q_{2k} \Rightarrow (\exists m')\big(m' \in P_{2k} \cap Q_{2k} \wedge 2k = m + m'\big)\,\big].$

Solutions come in pairs, stable under the involution $m \mapsto 2k - m$ whose fixed point is k — exactly the case GG22 handles. Goldbach decompositions go two by two.

Proof sketch, and what the migration cost. The partner of m is nothing other than the mirror's own internal witness y: it lies in S because the mirror says so, it is bounded because $y \leq m + y = 2k$, and m re-enters the mirror of y by commutativity of cardinal addition. No subtraction is used — there is none available — and no counting. The exploratory version of this proof was written with a *single* spelling of primality on both sets, which made the step invisible; on the repository's real definitions the two α-spellings cross, the partner leaving the mirror in one spelling and having to enter P in the other, while m makes the opposite trip. The bridge had to be parameterized by source and target spelling. A convenient simplification in an exploration script had concealed the entire real difficulty; the script "worked", it simply did not prove what we thought.

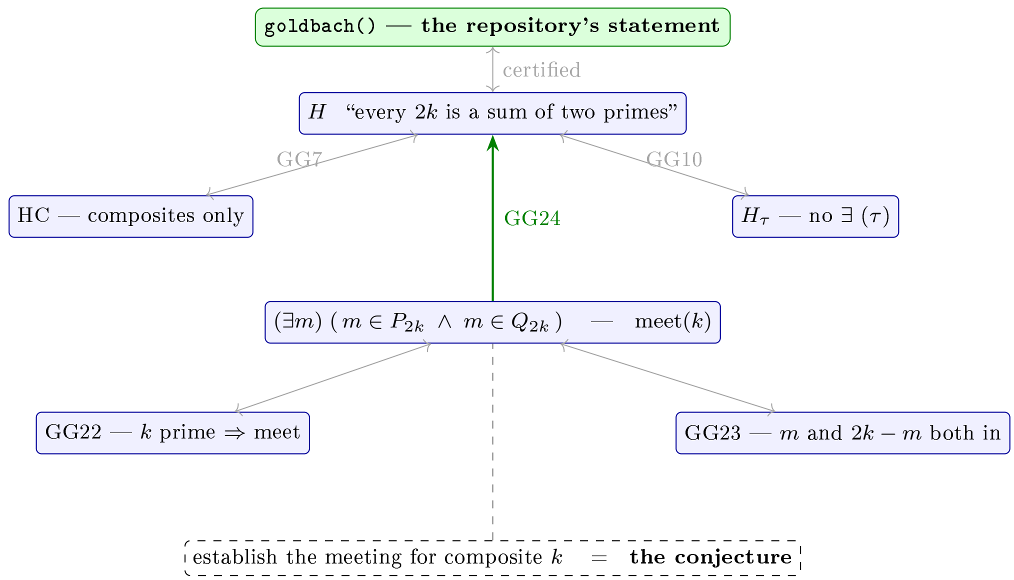

Figure 1: The map. Solid arrows are kernel-certified; the dashed link is the conjecture. P_{2k} is non-empty and finite (proved); Q_{2k} is its additive mirror. Every certified link is replayed by `capstone.py` and judged by the kernel.

Remark 2 (the negated bridge is not free). The contrapositive bridge $\neg\,\mathrm{prime}_2 \Rightarrow \neg\,\mathrm{prime}_1$ remains unavailable and does not follow from the parameterization — an implication does not contrapose for free here. The composites form carries the first spelling deliberately.

Proposition 4 (the half suffices). $\vdash (\forall k)\big[\, k \textit{ finite} \Rightarrow \big(\mathrm{meet}(k) \Leftrightarrow (\exists m)(m \in P_{2k} \cap Q_{2k} \wedge m \leq k)\big)\,\big]$.

The supporting lemma is pure cardinal arithmetic with no prime in it: if $2k = m + m'$ with all three finite, then $m \leq k$ or $m' \leq k$. Its proof avoids strict inequalities entirely — they are expensive in this kernel — by using comparability to open two cases, the existence of a complement in the case $k \leq m$ to write $m = k + d$, and then finite additive cancellation on $k + k = (k + d) + m' = k + (d + m')$ to get $k = d + m'$.

Remark 3 (the guard is not caution). Finite additive cancellation is *false* for infinite cardinals. Without the Fin guards this statement would not be merely unprovable — it would be wrong. This is the same class of defect as Section 7, caught in time rather than after the fact.

What this does not give, stated plainly. Halving a search space that remains infinite in k brings no proof closer. The conjecture is exactly as open as before. What is acquired is one more certified equivalence and one exact structural constraint: material for the map, not a step toward the solution.

Remark 4 (a methodological aside worth more than the result). The re-association step $(k + d) + m' = k + (d + m')$ uses iterated associativity of cardinal addition, which did not exist in the repository and had been proved that same morning on an unrelated track. A hole filled for one reason paid off for another within hours. It is the most concrete argument we have for closing gaps *when they are seen*, without waiting to need them.

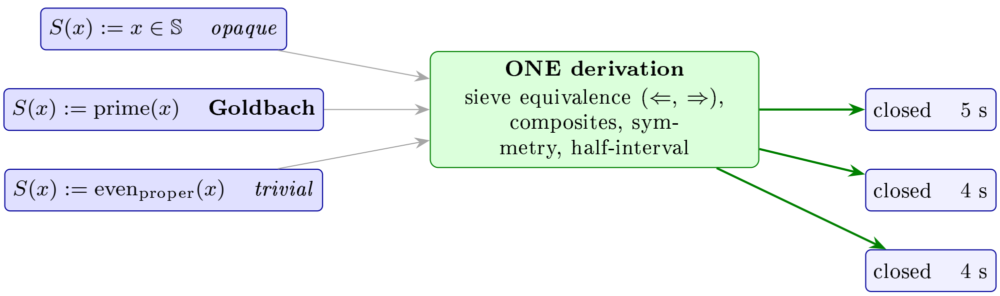

Figure 2: The measurement. One derivation, three predicates with nothing in common, three kernel-closed theorems. S is never opened: the derivation is not adapted per instance, it is *executed* with different arguments, and the kernel judges each result independently. A proof that cannot tell the primes from a structureless set cannot prove Goldbach — and that is a property of the statement it establishes, not a defect of the proof.

5 What the reductions do not contain

Sections 3 and 4 produce certified reductions of an open problem. The question that must eventually be asked of such a body of work is: *how much of the problem is in there?* We answer it by measurement.

5.1 The construction, with the predicate as a parameter

The module `recherche/additif/` rebuilds the sieve with the predicate abstracted — in the implementation, a function from terms to formulas:

$$P_b := \{\, x : \mathrm{Fin}\, x \wedge S(x) \wedge x \in [0,b] \,\}, \qquad Q_b := \{\, x : (\exists y)(\mathrm{Fin}\, y \wedge S(y) \wedge b = x + y) \,\},$$

and re-derives *all four* of the campaign's major reductions — the sieve equivalence in both directions, the reduction to composites, the symmetry, and the half-interval restriction — *without ever opening S*. The proofs are not adapted per instance; the same derivation is executed with different arguments.

S	time	what it is
$x \in \mathbb{S}$, $\mathbb{S}$ totally opaque	5 s	no property assumed whatsoever
est_premier(x)	4 s	**Goldbach**
est_pair_propre(x)	4 s	a set for which the question is trivial

The three rows are three tests in `tests/recherche/additif/`, named after what they establish: `test_symetrie_sur_un_predicat_totalement_opaque`, `test_goldbach_est_une_instance`, `test_un_predicat_sans_rapport_ferme_aussi`. Each asserts closure in the kernel sense and re-asserts the 22-axiom invariant. "Goldbach is an instance" is therefore an execution, not a sentence of prose.

5.2 The verdict

A proof that does not distinguish the primes from a structureless set cannot be used to prove Goldbach. This is not a defect of our proofs: it is a property of the statements they establish.

The symmetry of the sieve and the sufficiency of the half-interval carry *no arithmetic content* — they are facts about additive structure, and primality is an instance that plays no role in the derivation.

We regard this as a result rather than a confession, for three reasons. It *delimits* the conjecture, drawing the line between the structural and the arithmetic in code rather than by estimation. It *explains* the two paths closed negatively in Section 6: both fail for the same reason, never looking at which integers lie in S. And it is *cheap and repeatable* — a few seconds per instance — so it can be applied as a routine hygiene check to any reduction the campaign produces next, before that reduction is invested in.

Remark 5 (a side effect that confirms the reading). In the concrete development, the symmetry proof requires an α-spelling bridge in both directions, because the repository's statement forces two spellings of `est_premier`: the partner exits the mirror in one spelling and must enter P in the other. In the abstract version, with a single predicate, *that bridge disappears entirely.* It was therefore not a mathematical step but an artifact of notation — and it had cost real work.

5.3 How this measurement was itself found to be overstated

The paragraph above is the one we would most like the reader to distrust, so we report how it came to be true.

For a week, it was not. The campaign's internal map asserted that all four major reductions carry no arithmetic content; the code established *two* — the symmetry and the half-interval restriction. No parametric form of the sieve equivalence or of the composites reduction existed: a search of the parametric package for either returned nothing. The claim had been written in a module docstring on 12 August, propagated into the campaign's map document, and was on its way into this paper's central section.

It was caught on 19 August by reading the code against our own prose while drafting this section, and closed the same day: a new module derives the sieve equivalence in both directions and the composites reduction parametrically, closed in the kernel on all three predicates, with the parametric package's tests green (13 tests, 5 min 53 s) and the whole research tree re-run green (32 tests, 12 min 27 s). What the port revealed is the finding itself in a sharper form — **it was mechanical**. The concrete proofs use primality only as an opaque conjunct, carried from one end to the other and never opened; all the real work in them is cardinal arithmetic with no prime in it. There was nothing arithmetic to port, which is precisely what Section 5 claims.

Two things are worth extracting. First, the port *gained* generality: the abstract sieve equivalence holds for an arbitrary bound b, whereas the repository's version is written on $b = 2k$ and never uses that b is a double — only the composites reduction needs it, and only for a reflexivity. Second, and this is the methodological point: **no test catches a comment that lies about the code.** The kernel guarantees soundness; it guarantees nothing about the prose written beside it. This is the same class of defect as the fidelity failure of Section 7, one level up — there the formal statement drifted from the book, here the informal claim drifted from the formal statement. An unmeasured claim propagates faster than a measured one, and it propagates through exactly the channel a proof assistant does not police.

6 Two paths closed by the negative

Raw counting. The pigeonhole criterion $2 \cdot \pi(2k) > 2k + 1$ — which would force P_{2k} and its mirror to intersect by cardinality alone — holds for no $k \geq 2$. The density π/k falls from

0.50 to 0.18 by $2k = 2 \cdot 10^5$, while the actual number of decompositions *grows* (2 to 1417) over the same range. This is a numerical measurement, not a proof. Its consequence was a decision: do not formalize inclusion–exclusion for this problem. What is needed is information about the *distribution* of P_{2k}, not about its cardinality.

Equational reasoning. The map above was drawn by a machine that could not reason equationally — it built no congruences, chained no rewrites, instantiated no laws of the pool at encountered subterms. Since the core of Goldbach is an equation on cardinal sums, $2k = p + q$, the project's cycle (gap $\to$ organ $\to$ re-probe) required putting the obligations back in front of the new instrument. The commutations and the re-association close instantly. The generic obligation $\mathrm{meet}(k)$ does not, and — the point — the reported gap has *exactly the same shape* as before the three organs were built. The frontier did not move.

We are careful about what this shows. The two commutations were not out of reach beforehand: the journal records one of them closing in 0.0 s the day before. Only the re-association is new, and it is new because iterated associativity was promoted to the repository that morning — not because an organ improved. What the re-probe establishes is the negative: *Goldbach's obstruction is not equational*, so no strengthening of the algebra will touch it. The organs are real and useful, and strictly orthogonal to this conjecture.

One cause for both. Section 5 explains the pair. Counting looks only at $|S \cap [0, b]|$; equational reasoning looks only at the form of the statement. Neither ever inspects *which* integers are in S — and by the parametric measurement, neither can, since the reductions they operate on are valid for every S. What remains open remains open for the reason it did on the first day: one must produce an object — a prime $m \leq 2k$ whose complement $2k - m$ is prime — and no formal manipulation of the statement fabricates one.

7 Fidelity: a defect in the statement of primality

The kernel guarantees soundness — no false theorem — but not *fidelity*: that the formalized statement is the one intended. This campaign produced a clean instance, certified rather than argued.

The repository's `est_premier`(p) guards only the *divisor*: since $\mathrm{divides\_properly}(d, p)$ requires $p = \mathrm{Card}(d \times q)$, an object p that is not a cardinal is divisible by nothing at all, the universal clause is vacuously true, and "prime" collapses to $p \neq 1$. In the kernel:

$$\vdash \big(p \neq 1 \;\wedge\; (\forall d)\, \neg\, \mathrm{divides\_properly}(d, p) \big) \implies \mathrm{est\_premier}(p).$$

Everything $\neq 1$ that nothing divides is "prime". Consequently `goldbach()` quantified over witnesses not required to be integers: **the formalized statement was strictly weaker than the conjecture**. Soundness was never at risk; fidelity was.

The repair is $\mathrm{prime\_int}(p) := \mathrm{Fin}(p) \wedge \mathrm{est\_premier}(p)$, and its cost is certified to be zero where it matters — a kernel theorem establishes that the guard is free on numerals, i.e. that we lose nothing on the objects the campaign actually computes with. The guarded arc implies the repository's historical statement (link GG21); the converse is not proved, and the audit module says so explicitly.

Two lessons are on record. First, the same fault — a quantifier ranging over sets while believed to range over integers — had been committed the previous day on primality's $(\forall d)$: it is a *pattern*, not an accident. Second, and more useful: the *proved* bounded variant of the conjecture had carried the finiteness guard from its first draft, because its proof demanded it; only the *unproved* statement had drifted. Proving forces hypotheses to be named. Stating forces nothing.

8 Reproducibility, and the two guards

The arc is not a pile of session scripts. It is a permanent package of nine modules under `recherche/goldbach/`, plus the parametric package of Section 5 under `recherche/additif/`, with 32 tests across the research tree — all green, 12 min 27 s, re-run 19 August 2026 — and `capstone.py` replays the chain with the kernel as sole judge. Re-run in a fresh process on 19 August 2026: **18 links, all closed**, `theorie_ensembles()` = 22, about seven and a half minutes end to end. The dominant costs are the half-interval lemma (259 s) and its sieve instance (120 s); the guard-is-free theorem takes 50 s; ten links are instantaneous.

Guard one: "closed" does not mean "axiom-free". A theorem drawn by the `axiom` rule has an empty hypothesis set, so closure says nothing about the dedicated theories a derivation rests on. The sieve has two axioms characterizing P_b and Q_b. Conflating the two notions is the only real cheating available in this setting, so the capstone carries a dedicated column: **eleven of the eighteen links are free of ad-hoc axioms; seven rest on the sieve's two**. Every affected function passes through an attestation helper that prints them.

Guard two: a test that fails on good news. A test (`test_goldbach_reste_ouverte`) sweeps every export of the research package and fails if any of them concludes H — the conjecture — on its own. Everything the package produces must remain of the form $X \Rightarrow$ Goldbach or Goldbach $\Leftrightarrow Y$. If that test ever goes red, it is not a result to publish: it is a statement to audit immediately. We consider this the single most important line of the package, because the failure mode of a campaign like this one is not unsoundness — the kernel prevents that — but a statement that has quietly drifted into vacuity.

What was abandoned, and why. Results built on an *unguarded* bounded prime set were not migrated. They concern a mathematically different set from the guarded one, the guard being precisely the correction of Section 7. Re-introducing them would require re-proving them on the guarded set. We record this rather than quietly carrying the old numbers forward.

9 Related work

What is known about Goldbach, and what we do not add to it. The even conjecture has been verified numerically for all $n \leq 4 \cdot 10^{18}$, together with extensive data on minimal partitions and prime gaps [7]. The ternary conjecture — every odd $n > 5$ is a sum of three primes — was proved by Helfgott [3], resting on a numerical verification up to $8.875 \cdot 10^{30}$ [4]. Against that background our bounded arc, which certifies decompositions for $n \leq 86$, is arithmetically negligible, and we do not present it as a contribution: its role is structural, feeding the concrete witnesses that GG25 absorbs into the meeting form. We add no arithmetic fact, and nothing here bears on the ternary case. To our knowledge neither Helfgott's proof nor the numerical verifications have been mechanized, and we do not attempt either.

Formalizing statements that have no proof. Large collaborative formalizations of recent mathematics — the Polynomial Freiman–Ruzsa project is the best-documented example, with a human-readable blueprint linked to the formal development [9] — take as input a proof that *exists* and transcribe it. The activity reported here is different in kind: there is no proof to transcribe. What gets formalized is the statement, its equivalent forms, and the exact obligation at each leaf. The two are complementary rather than competing, and the distinction matters for what one can promise: a blueprint project can be finished, whereas a map of an open problem can only be extended or shown, as in Section 5, to be measuring less than it appeared to.

Measuring what a theorem needs. The closest conceptual relative to our central result is reverse mathematics, which calibrates theorems by the subsystem of second-order arithmetic required to prove them [8]. The question is the same in spirit — *what does this proof actually use?* — and the answers have the same negative flavour: a theorem provable in a weak system cannot, by itself, yield one that is not. Two differences are worth stating. Reverse mathematics calibrates against a fixed hierarchy of set-existence axioms; we calibrate against a *parameter in the statement*, asking whether a specific predicate is used at all. And our calibration is executable rather than proof-theoretic: it consists in running the same derivation with different arguments and letting the kernel judge, which takes seconds and can be applied routinely to a reduction before one invests in it. We do not claim a reverse-mathematical result; we claim a cheap, mechanical instrument in the same family.

Negative results in automated reasoning. Systems that establish something by failing to prove — counterexample search [1], and more recently learned disproof [5] — have a long history, and the survey literature on conjecture generation catalogues what such systems propose [10]. Our negative results are of a different type. They do not refute a statement; they establish that a family of *our own* statements is insensitive to the parameter the problem turns on. Nothing is disproved, and the conjecture's truth value is untouched.

Bourbaki in a machine. Prior mechanization of *Théorie des ensembles* formalizes the content over the host logic rather than inhabiting Bourbaki's own formalism [2], for reasons the size of τ-expanded assemblies makes plain [6]. The kernel used here inhabits the formalism, which is what makes the τ rewriting of Section 3 a kernel operation rather than an encoding trick; the companion paper argues that point and we do not repeat it.

Novelty, honestly stated. Formalizing an open conjecture is not new, and neither is reducing one to an equivalent form. What we have not found elsewhere is the *measurement* of a reduction's arithmetic content by executable parameterization: running one derivation against an opaque predicate, against the intended one, and against a trivial one, and reporting that all three close. Our claim of novelty is semi-decidable in the sense the companion paper insists on — no counterexample after methodical search, never more — and it is a claim about the *instrument*, not about Goldbach.

10 Limitations and threats to validity

The measurement covers four reductions, not the whole map. All four major reductions are now measured (Section 5.3), but "major" is our judgement. Smaller links of the chain — the τ

bridge, the bounded arc absorbed by GG25 — have not been re-derived parametrically, and we do not claim they are free of arithmetic. The honest statement is that every reduction we identified as load-bearing has been measured, not that the entire eighteen-link chain has.

The negative results are measurements. The counting result is numerical evidence about π over a finite range, not a theorem; a proof that no counting-only argument can succeed is not offered. The equational result describes the shape of a gap reported by our organs at a given state of the system, and a different instrument could in principle report differently.

Two dedicated axioms. Seven of the eighteen links rest on the sieve's axioms characterizing P_b and Q_b. They are bounded selections, deliberately built to avoid the incoherence class the project met in July, but they are assumptions and they are counted as such.

One mind. The system was designed, operated and audited by a single person. The guards of Section 8 exist precisely because self-audit is weak, but they do not replace external review.

An exact oracle. Everything here enjoys a decision procedure for "is this a theorem". The methodology — parameterize, re-derive, report what still closes — should transfer to any system with a checkable kernel, but we have not tested it outside this one.

11 Conclusion

We set out to see what a kernel-verified system produces on a problem it cannot solve. It produced a map: eighteen certified links collapsing three lines of attack onto one object, a conjecture rewritten without existential quantifiers, two exact structural constraints on the solution set, and a fidelity defect in the statement itself, proved and repaired. None of it advances the conjecture by a step, and we say so at every turn.

The result we did not anticipate is the one we would keep. By replacing primality with a parameter and re-running the derivations unchanged, we measured that our own reductions contain no arithmetic — and thereby explained, at one stroke, why two independent lines of attack had already failed. A research program cannot easily tell, from the inside, whether it is approaching a problem or orbiting it. Here that question became a test that runs in four seconds and reports an answer we did not want.

That, we think, is the transferable object: not the map, but the habit of asking a verified system to measure the content of its own reductions — and being willing to publish the measurement when it says the work was structural all along. Goldbach needs information about *which* integers are prime. Everything above is the precise, certified statement of the fact that we have not supplied any.

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